Fiscal Year:	FY 2016	**Task Last Updated:**	FY 07/29/2015
PI Name:	Kelton, Kenneth F. Ph.D.		
Project Title:	NASA Research under ESA-Based Investigations THERMOLAB and ICOPROSOL		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	10/01/2010	**End Date:**	12/31/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 12/31/2015 per NSSC information (Ed., 7/8/15) NOTE: To be extended to 9/30/2015 per PI (Ed., 7/10/13)		
Key Personnel Changes/Previous PI:	Kenneth Kelton is U.S. Team Member on ESA-sponsored projects, “Thermophysical properties and solidification behaviour of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL)," and "Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS)."		
COI Name (Institution):			
Grant/Contract No.:	NNX10AU19G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Funding is for Dr. Kelton's role as Team Member on the subprojects described below, associated with European Space Agency (ESA)-sponsored main projects as noted. Main project: Thermophysical and structural properties and solidification behavior of supercooled glass forming liquids as a member of THERMOLAB; the Science Coordinator is Prof. Dr. Hans Fecht, Ulm University, Institute of Micro and Nanomaterials, Albert-Einstein-Allee 47, Ulm Germany. Similar measurements on undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL); the Science Coordinator is Dr. Dirk Holland-Moritz, Deutsches Zentrum für Luft- und Raumfahrt (DLR), Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany. Subprojects: 1. Ground-Based Experimental Program: Synchrotron X-ray studies of liquid structures are made at the Advanced Photon Source using the ESL (electrostatic levitation) chamber at Washington University (WU-BESL), which has been optimized for scattering studies. These are correlated with thermophysical property data obtained on ground with WU-BESL and using the Material Science Laboratory Electromagnetic Levitator (MSL-EML) on the International Space Station (ISS). Terrestrial maximum undercooling and induction time to recalescence data are obtained in ESL, where stirring ensures that nucleation is not diffusion controlled. This data will allow an evaluation of diffusion in the upcoming MSL-EML ISS undercooling studies. 2. Thermophysical Property Measurements: ISS. Specific heat, density, surface tension and viscosity measurements will be made of supercooled Ti-Zr-Ni liquids with different Ni concentrations as a function of the temperature using the MSL-EML facility. These will be correlated with x-ray synchrotron diffraction studies of the supercooled liquids and ground-based measurements made using WU-BESL The purpose is to determine the existence and nature of liquid/liquid phase transitions that may be associated with chemical and topological ordering and a possible connection with liquid fragility. 3. Effect of diffusion on nucleation: The importance of long-range diffusion on nucleation will be evaluated using the MSL-EML facility located on the ISS. The critical items of interest are the maximum undercooling and the induction time to recalescence in isothermal holds at temperatures below the liquidus temperature. Together with the results from the thermophysical property measurements and an MHD convection model results this will allow us to analyze how the maximum undercooling is influenced by fluid flow. These results of the micro-g experiments will be compared with the results of ground-based ESL measurements and with predictions from Kelton's coupled-flux model for nucleation. Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany Subprojects: 1. Ground Based Experiment Program. Measurements of the density, surface tension and viscosity and synchrotron x-ray diffraction measurements of the liquid structure in the supercooled state using WU-BESL, Ground-based measurements of the maximum undercooling and induction time to recalescence in isothermal holds will also be made using WU-BESL. These will be used to evaluate the nature of ordering on phase transitions in the supercooled liquid and fragility. 2 Long Duration Microgravity Experiments. Measurements of thermophysical properties, specific heat (modulated measurements), surface tension, viscosity and density, and also electrical resistivity will be made as a function of temperature using the MSL-EML on the ISS. These will be coupled with x-ray and elastic neutron diffraction measurements of the liquid structure factor (WP 2) and ground based studies of the density and viscosity to study the role of ordering on novel liquid/liquid phase transitions and to determine the nature of these transitions. Maximum undercooling and time to recalescece measurements as a function of temperature and stirring will be compared with terrestrial ESL measurements to assess the importance of diffusion on nucleation, allowing Kelton's coupled flux model for nucleation to be evaluated experimentally.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Product advancement in industries such as information technology, aerospace, automotive, and basic and new materials manufacturing requires advanced materials development. In the past, this was often carried out through a laborious process of guided trial and error. The vast increases in computing power have now opened the door for realistic modeling approaches, which has the potential to greatly improve the efficiency of materials development. In the US, this has become a central focus of research in materials science, encapsulated in the Materials Genome Initiative. However, progress is frequently hampered by the lack of relevant thermophysical property data needed for the modeling. Further, key processes such as crystal nucleation from the liquid or other phase transitions within the liquid, are often poorly understood, making it difficult to realistically model them. In many cases, the experimental data and a deeper understanding of some processes cannot be obtained from terrestrial experiments due to interactions with sample containers at the temperatures of interest. Even containerless based methods, such as electrostatic levitation, cannot be used to accurately investigate these. The ISS experiments planned under the support of this grant will allow these data to be obtained.
	NNX10AU19G is focused on flight and ground-based studies of properties of and phase transitions in supercooled high-temperature metallic liquids. The acronym for the US team funded under this grant is QUASI. This grant supports QUASI's flight opportunity on an ESA facility, working in collaboration with two European teams (THERMOLAB and ICOPROSOL). The focus of NNX10AU19G is to support ground-based structural, density, surface tension and viscosity studies of the supercooled liquids, and to support ISS-based studies of specific heat, emissivity, thermal conductivity, viscosity, and nucleation studies in a micro-gravity environment. The goal of the research is to develop a deeper understanding of the mechanisms of nucleation and growth during liquid crystallization, understand the relation between the atomic structure of the liquids and their physical properties, and to investigate possible liquid/liquid phase transitions. This past year, NNX10AU19G has provided partial support for the Principal Investigator (PI), a Research Scientist, and two graduate students. The key activities and results of ground-based studies over this period are listed below.

Task Progress:

A. Upcoming Studies on the International Space Station (ISS). The flight experiments on the ISS are designated as Batches. Samples have been selected for Batch 1 and Batch 2 and preliminary planning for Batch 3 has started. The status of the ISS planned studies follows below.

Batch 1 - Investigations of samples in Batch 1 were initiated in Spring 2015. As a formal member of the THERMOLAB team, QUASI will study the liquid that forms the deeply metastable metallic glass, Vit 106a. QUASI is the team leader for a similar sample in the same batch, Vit 106. The focus of these studies will be primarily on thermophysical property measurements and the possible presence of a liquid/liquid phase transition. Measurements of the maximum undercooling before crystallization will also be made as a function of stirring to investigate diffusion effects on the nucleation step of crystallization, which requires a quiescent microgravity environment that cannot be attained under terrestrial conditions. Accurate measurements of the specific heat, a key thermodynamic parameter, are also possible using the electromagnetic levitation (EML) facility on the ISS. These are not possible terrestrially due to the large densities and low melting temperatures of the samples of interest, making it impossible to supercool the samples in EML while maintaining levitation. Experimental studies of Vit 106a and Vit 106 are expected to begin in mid to late Fall 2015. Additionally, QUASI has formal interests in thermophysical property studies of several other samples in Batch 1.

Batch 2 - The selection of Batch 2 samples were recently completed. As a member of the THERMOLAB team, QUASI will participate in studies of Zr52.5Cu17.9Ni4.6Al10Ti5 (LM105), another good glass forming liquid, again focusing primarily on thermophysical property measurements, but also studying the maximum undercooling as a function of stirring. QUASI is the team leader for thermophysical property and maximum undercooling studies of Zr47Cu47Al6, another bulk metallic glass-forming alloy. As a member of ICOPROSOL, QUASI will participate in studies of Ti39.5Zr39.5Ni21, focusing primarily on nucleation and growth studies to investigate the role of diffusion in crystallization. QUASI has formally stated interests in other samples, in particular Zr64Ni36 and Cu50Zr50, for which QUASI has obtained extensive structural and chemical ordering data in the supercooled liquids that will allow the relations between structure and physical properties to be explored.

Batches 3 & 4 – Initial planning by the Internal Working Group for samples on Batches 3 and 4 is starting. At this time, QUASI tentatively has one sample in each batch, Zr80Pt20 (Batch 3) and Zr75.5Pd24.5 (Batch 4). The currently planned primary focus for these samples will be measurements of thermophysical property measurements, in particular specific heat, which will be used with ground-based X-ray and elastic neutron scattering data to probe the observed differences in what should be similar alloy liquids, based on atomic sizes and heats of mixing.

B. Activities over the Past Year. The key activities over the past year are listed below:

1. Supported three Internal Working Group meetings at the DLR (German Aerospace Center) in Cologne, Germany: October 1-2, 2014, January 8, 2015 (joined remotely) and May 11-12, 2015.

2. Supported a visit to MUSC at the DLR from May 6-8, 2013 to prepare the Science Protocol Documents for Ni-Zr and Ti-Zr-Ni for Batch 2 experiments and for Cu-Zr-Al (Batch 3).

3. Calorimetric studies of liquid fragility and ground-based specific heat measurements, in anticipation of thermophysical property measurements on the ISS.

4. Identification of a possible correlation between liquid density and glass formability.

5. Preparation Cu47Zr47Al6 and Zr80Pt20 (ISS Batch 3) samples for parabolic flight experiments and parameter definition and analyzed the viscosity measurements from the flight experiment.

6. Measured evaporation rates for Zr80Pt20, a sample for Batch 3.

7. Dust and coal coating calculations for ISS samples Zr-Ni and Ti-Zr-Ni (ISS samples in Batch 2) and Zr-Cu-Al (ISS sample in Batch 3).

C. Brief Summary of Three Key Results from Ground-Based Activities. All of the points listed above are discussed in detail in the annual report available to NASA management. Three key results are discussed in brief below.

1. Calorimetric methods for determining glass and liquid fragility – Liquid or glass fragility is a key concept used to classify liquids. Liquids are classified as “strong” when the logarithm of the viscosity as a function of inverse temperature has a constant slope (activation energy). They are “fragile” when the slope of this plot becomes increasingly larger on approaching the glass transition temperature (defined as the temperature where the viscosity attains the value of 10 to the power 12 in units of Pa.s). Fragility is generally accepted as being a good predictor of glass formability, making it of practical as well as basic importance. While viscosity data are often lacking, studies have shown that fragility can be estimated from calorimetric data, frequently taken with a commercially available Differential Scanning Calorimeter (DSC). In support of the upcoming viscosity and other themophysical property studies on the ISS, we have evaluated the most commonly used methods for obtaining fragility from DSC data. We have demonstrated that while most work reasonably well for strong liquids, which easily form glasses, they do not work well for fragile liquids, which are generally poor glass formers. We have developed a method that works for these latter glasses, based on the temperature scan rate dependence of the onset of crystallization.

2. Method for estimating the total hemispherical emissivity by a combination of Calorimetric and Electrostatic Levitation (ESL) measurements – For accurate temperature measurements of liquids during containerless processing, and to determine the temperature-dependent specific heat, it is essential to know the total hemispherical emissivity for the liquid samples. Values for this quantity, however, are generally lacking, due to the difficulties of terrestrial measurements of levitated liquids. We have developed an ESL-based method using the heat of fusion, determined calorimetrically, that allows the temperature dependent emissivity to be extracted from ESL measurements by balancing the radiation emitted from the levitated samples with the input laser power. The values obtained using the ESL will be compared with those obtained more directly using the EML on the ISS to evaluate the accuracy of the new method.

3. Correlation of liquid density and glass formability – From ESL-based measurements of the containerlessly-processed liquid samples, we have shown that both the rate of change of the volume with temperature and the volume change upon melting are correlated with glass forming ability. The volumes of the liquids for better glass formers (stronger liquids) change less on cooling from a high temperature near the melting temperature to the glass transition temperature. Further, the volume change upon melting is less for strong liquids, suggesting that these liquids have compact low energy structures compared with those of more fragile liquids. Taken together all of our studies demonstrate that glass formability and liquid fragility are embedded in the high temperature liquids, in agreement with our studies reported last

	year of the existence of a high temperature near the melting temperature where cooperative flow first sets in and which eventually results in the glass transition. [Editor's note (February 2016): project continues as "NASA Research under ESA-Based Investigations THERMOLAB and ICOPROSOL," grant number NNX16AB52G, with period of performance of 1/1/2016-12/31/2020.]
Bibliography Type:	Description: (Last Updated: 05/20/2025)
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Abstracts for Journals and Proceedings	Kelton KF. "Fragility in Metallic Liquids." Invited presentation in Symposium UU: Structure-Property Relations in Amorphous Solids, 2014 Fall Meeting of the Materials Research Society, Boston, MA, November 30 - December 5, 2014. 014 Fall Meeting of the Materials Research Society, Boston, MA, November 30 - December 5, 2014. Abstract book distributed at the meeting. Abstract # UU4.02. , Dec-2014
Abstracts for Journals and Proceedings	Kelton KF. "Kinetic and Structural Fragility in Metallic Glass-Forming Liquids." Plenary lecture in the AIMR International Symposium (AMIS2015), Sendai, Japan, February 17-19, 2015. IMR International Symposium (AMIS2015), Sendai, Japan, February 17-19, 2015. Abstract book distributed at the meeting. , Feb-2015
Abstracts for Journals and Proceedings	Kelton KF, Blodget ME, Johnson, MA, Mauro, NA, Vogt A. "Link between Liquid Fragility and Structural and Chemical Ordering." Invited presentation at ISMANAM 2015 (International Symposium on Metastable, Amorphous and Nanostructured Materials), Paris, France, July 13-17, 2015. ISMANAM 2015 (International Symposium on Metastable, Amorphous and Nanostructured Materials), Paris, France, July 13-17, 2015. Abstract book distributed at the meeting. , Jul-2015
Abstracts for Journals and Proceedings	Kelton KF. "Correlation between Kinetic and Structural Fragility in Metallic Glasses." Invited presentation at a CECAM (Centre European de Calcul Atomique et Moleculaire) workshop, The Role of Structure in Dynamical Arrest, Schloss Waldthausen, Mainz, Germany, July 21-23, 2015. CECAM (Centre European de Calcul Atomique et Moleculaire) workshop, The Role of Structure in Dynamical Arrest, Schloss Waldthausen, Mainz, Germany, July 21-23, 2015. Abstract book distributed at the workshop. , Jul-2015
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